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Online Shopping System

Module 5 UML Use Case Diagram

OVERVIEW

This use case diagram shows what an online shopping system does for the people and outside systems that use it. Everything inside the box labeled Online Shopping System is part of the system itself. The actors sit outside the box, since they either start a use case or receive the results of one.

I kept the main shopping path (view products, add to cart, make a purchase, check out) running down the middle so it reads in roughly the order a customer would actually go through it. The diagram was built in draw.io using the UML shapes.

ACTORS

Primary actors

These are the people who start use cases on their own.

- **Guest.** A shopper who is not signed in. A guest can register a new account, view products, and add items to a cart.
- **Registered Customer.** A shopper with an account. The hollow arrow means a registered customer is a kind of guest, so they can do everything a guest can, plus log in, make purchases, and track orders.
- **Administrator.** Store staff who log in to manage products and orders.

Secondary actors (outside systems)

These are on the right side because they never start a use case. The store calls on them to get something done.

- **Email Service** sends account and order emails.
- **Payment Gateway** charges the customer's payment during checkout.
- **Shipping Provider** reports shipping status for Track Order.

USE CASES

The main use cases are Register New Account, Log In, View Products, Add Items to Cart, Make Purchase, Checkout, Track Order, Manage Products, and Manage Orders. The green

ovals are the main use cases and the yellow ovals are the supporting ones that get pulled in through «include» or added through «extend».

RELATIONSHIPS

«include»

An included use case always runs as part of the base use case. The dashed arrow points from the base to the use case it includes.

«extend»

An extending use case is optional and only happens under certain conditions, so its arrow points back from the optional use case to the base it adds to.

Type	Base use case	Connected use case	Why
«include»	Register New Account	Send Email Notification	Every new account gets a confirmation email
«include»	Log In	Verify Credentials	Logging in always checks the username/password
«include»	Make Purchase	Checkout	A purchase cannot finish without checking out
«include»	Checkout	Process Payment	Checkout always charges the payment
«include»	Checkout	Send Email Notification	Order confirmation (shared with registration, so it is only modeled once)
«extend»	View Products	Search Products	Optional, a shopper can browse without searching
«extend»	View Products	View Product Details	Optional, only when a shopper opens a product
«extend»	Checkout	Apply Coupon Code	Only when the customer has a coupon
«extend»	Log In	Reset Password	Only when the user forgets their password

USE CASE DIAGRAM

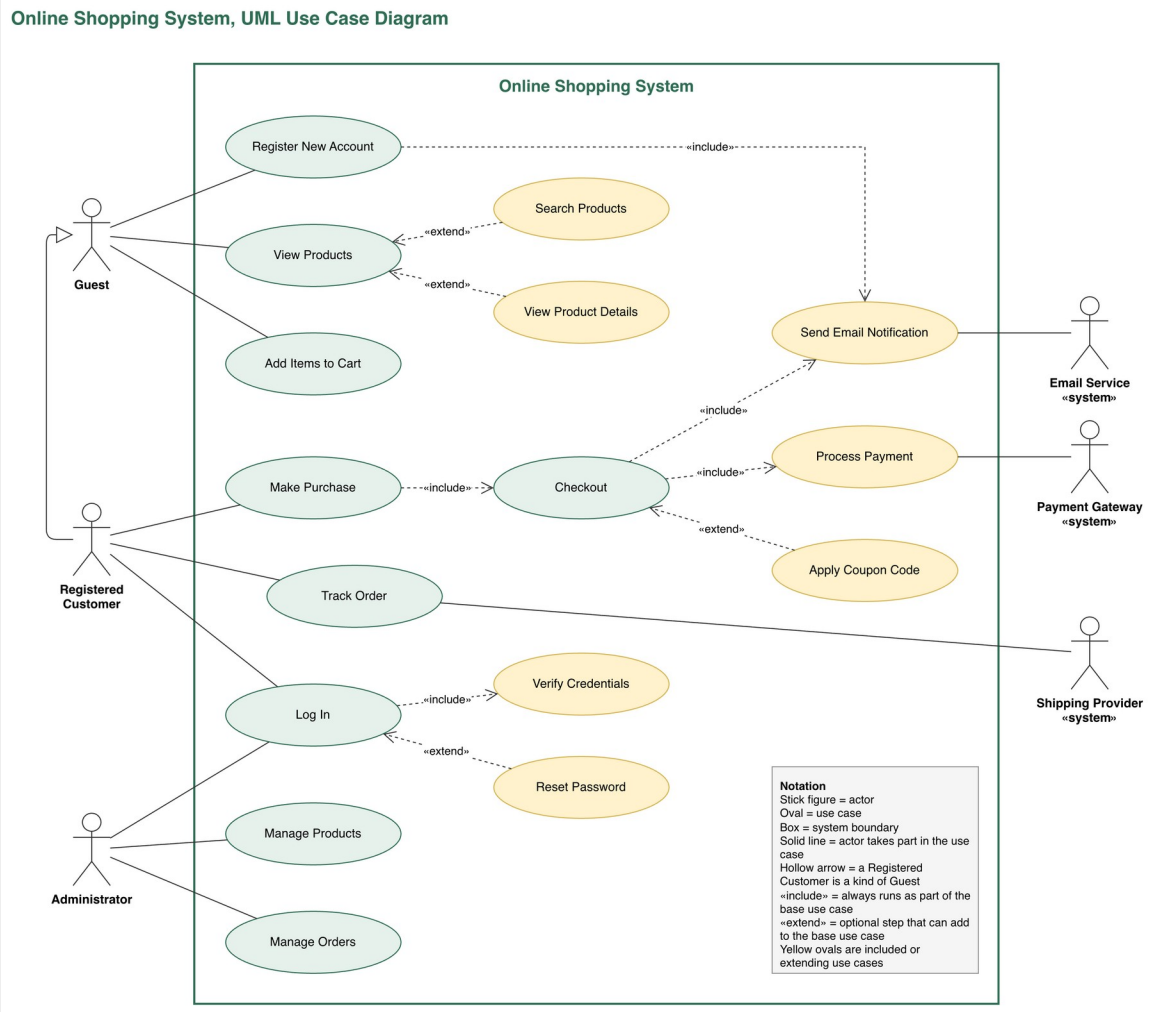


Figure 1 | Online shopping system use case diagram with actors, the system boundary, «include», and «extend»